PART FIVE

The Evidence

Crop by Crop

CHAPTER THIRTY-THREE

Rice as Proof of Concept

2008 to Present, Peer-Reviewed Foundation, Field Replication, and Scale

"Rice is one of agriculture's most water-intensive staple crops. The 2009 raised-bed trial, with a hybrid coarse rice, produced 12.84 tonnes per hectare, compared with a regional average of about 4 tonnes. Later mature PQNK field records of Basmati used about 8 percent of the water of the flooded-rice reference. The two results are separate, but together they make rice the system's most demanding proof of concept."
Asif Sharif, Lahore, 2026

Part Five tests the science and protocol developed in Parts One through Four against crop-specific evidence. Each chapter separates peer-reviewed findings from later PQNK field records and compares results only where the season, soil, climate, and baseline permit a valid comparison. The purpose is to show what PQNK changes in the root zone, crop, water use, yield, and farm accounts.

Rice begins Part Five because rice was PQNK's first deliberately designed experiment, its most challenging test, its first peer-reviewed publication, and its most globally significant result. No crop makes the case for PQNK more completely than rice, because no crop violates all four of PQNK's governing rules as thoroughly as conventional paddy does, and no result demonstrates more clearly what biological restoration produces when those violations are removed.

THE GLOBAL CONTEXT: RICE, WATER, AND THE COMING CRISIS

Rice feeds more than half of humanity and is a primary staple across Asia, sub-Saharan Africa, and parts of Latin America. It occupies roughly 160 million hectares worldwide. Its water demand is especially important in Asia: IRRI research estimates that irrigated rice receives 34 to 43 percent of the world's irrigation water, or about 24 to 30 percent of its developed freshwater resources and occupies a much larger share of irrigated land in several Asian regions.

Population growth, declining aquifers, and greater climate variability make flooded paddy one of agriculture's most consequential production challenges. Rice does not biologically require continuous flooding, yet much of the crop is managed in saturated or flooded soil. That practice consumes large volumes of water, produces methane, suppresses aerobic soil functions, and confines roots to a shallow working zone. The problem lies in the production environment, not in the rice plant.

The PQNK rice system addresses these linked constraints. The founding raised-bed trial, published in a peer-reviewed journal in 2011, grew a hybrid coarse rice and produced more than three times the regional yield while using about 70 percent less irrigation water. Later mature PQNK fields of Basmati, on large acreages across many farms, yield at least 20 percent more than conventional Basmati grown alongside them, with no purchased inputs

and about 8 percent of the water of a flooded-rice reference. Because PQNK fields are not flooded, they remove the persistent anaerobic conditions responsible for most paddy methane. This chapter presents the 2009 experiment first, the mature system second, and the subsequent field experience separately.

BEFORE THE FIRST TRIAL: TWO QUESTIONS ABOUT RICE

When we were planning the first trial in 2009, to grow rice in moist soil rather than in flooded, saturated soil, we first studied two questions. Why is rice grown in standing water? And why is it mostly transplanted from a nursery rather than sown directly in the field?

Why rice is grown in standing water. Rice is a grass. It adapted to survive in low-lying land where monsoon rain collected and stood for weeks, where other plants drowned. Because rice survives where other crops cannot, it came to be believed that rice likes to grow in flooded soil. That belief was proved wrong when rice was planted on raised beds in moist soil and thrived.

We later learned how the rice plant survives flooding. Roots need oxygen, and waterlogged soil holds almost none, because oxygen moves through water about 10,000 times more slowly than through air. Rice responds by building its own air supply. Cells inside its roots, stems and leaf sheaths break down to leave a network of open air channels, called aerenchyma, running from the leaves to the root tips. Oxygen taken in above the water travels down these channels to the roots. On the older parts of each root a tight outer layer forms, so that the oxygen is not lost into the soil before it reaches the growing tip. A little oxygen escapes near the tip and keeps a thin layer of soil around the root free of the iron and other toxic compounds that build up in waterlogged soil.

This is a survival mechanism, not a sign that rice prefers water. Building and maintaining these air channels costs the plant energy, its roots stay in a shallow zone, and the plant lives under constant stress. When the same plant was grown in moist, aerated soil, free of that stress, it rooted deeply, tillered strongly and thrived.

Why rice is transplanted. The tradition of transplanting comes from a time before canal irrigation and groundwater pumping, when rice was grown in the monsoon season on rain alone. Before the rains begin the soil is dry, and once water stands in the field, seed will not germinate under it. Farmers therefore raised a nursery on a small patch, watered by hand pump, well before the rains, and began transplanting as soon as water ponded in the fields. It was the only practical way, and it is still largely followed around the world.

PQNK has changed this understanding. On permanent raised beds kept moist and never flooded, rice needs neither standing water nor a nursery. In the mature PQNK system it is direct-seeded into the beds, and it yields more with far less water.

THE CORE DATA: THE FOUNDING TRIAL AND THE MATURE SYSTEM

The founding experiment was conducted in 2009 on newly formed raised beds in Punjab and is described in Chapter Nineteen. Its results were published in 2011. The mature PQNK rice system developed later: rice was direct-seeded into permanent mulched beds

and managed without flooding. These evidence records must remain separate. The founding trial used a hybrid coarse rice, transplanted seedlings, fertiliser, and compost; it was neither a zero-input trial nor the source of the 92 percent water-saving figure. The mature record is of conventional Basmati varieties grown with no purchased inputs. Because the varieties and inputs differ, the yields of the two records are not compared with each other; each is compared only with conventional rice of the same variety.

The founding trial, as published:

Metric	Conventional flooded paddy	Founding raised-bed trial (2009)	Change	Source
Yield	About 4 t/ha regional average	12.84 t/ha crop cut from 8-ha block	More than 3x	Sharif 2011
Irrigation water	Continuous flooding	Furrow irrigation; no standing water after establishment	About 70% less	Sharif 2011
Labour	Manual transplanting and flood management	Water-wheel transplanter; mechanical weeding	About 70% less	Sharif 2011
Establishment	Nursery seedlings into puddled paddy	10-day-old seedlings; 22.5 cm square spacing	Different method	Sharif 2011
Nutrition	Broadcast fertiliser	Banded fertiliser plus compost	Targeted; not zero-input	Sharif 2011
Field form	Puddled flat field	Newly formed raised beds	Different system	Chapter Nineteen

Peer-reviewed source: Sharif, A. (2011). Technical adaptations for mechanized SRI production to achieve water saving and increased profitability in Punjab, Pakistan. Paddy and Water Environment, 9(1), 111–119. https://doi.org/10.1007/s10333-010-0223-5

The mature PQNK rice system, from later field records:

Metric	Flooded-rice reference	Mature PQNK field record	Change	Source
Crop-life water	6,579,200 L/acre	About 513,950 L/ acre	About 92% less	PQNK field record
Water per kg paddy	Not stated	About 321 L at 1,600 kg/acre	Separate field record	PQNK field record
Establishment	Transplanting into puddled paddy	Direct seeding into permanent mulched beds	Transplanting removed	PQNK protocol
Tillage and puddling	2–3 operations per season	None after conversion	Eliminated	PQNK protocol
Purchased inputs	Routine fertiliser and sprays	No purchased inputs after field maturity	Eliminated at maturity	Chapter Thirty-Two
Methane	High under continuous flooding	Much lower; beds never submerged	Anaerobic source removed	PQNK protocol

The mature-system water record was obtained at 1,600 kg of paddy per acre, the average Basmati yield across large PQNK acreages on many farms. Side by side with conventional Basmati of the same variety in the same season, PQNK yields at least 20 percent more, with no purchased inputs and about 92 percent less irrigation water. The 12.84 t/ha founding yield and the 92 percent water saving come from different records, different varieties and different inputs, and must never be presented as one combined result.

THE BIOLOGICAL EXPLANATION: WHY PQNK RICE OUTPERFORMS

The 12.84 t/ha yield of the founding hybrid crop shows what became possible when root-zone aeration, plant spacing, water management, and timely field operations were changed together. The mature Basmati record shows the same mechanisms at work at scale, at least 20 percent above conventional Basmati.

Aerobic root-zone conditions support deeper rooting. In flooded paddy, oxygen shortage and the compacted puddled layer restrict root development. In the founding trial, rice roots in aerated soil above a fractured hardpan were observed at depths of 24 inches or more. That larger rooting volume improved access to water and mineral nutrients and helped support grain filling.

Precision spacing supports tiller development. At 22.5-centimetre, or about 9-inch, square spacing, each plant had sufficient root-zone volume and light for primary, secondary, and tertiary tillers to develop. The response came from the interaction of spacing with aerated soil, controlled moisture, and reduced competition; spacing alone cannot reproduce the result.

Aerated soil strengthens biological nutrient delivery. Flooding does not eliminate arbuscular mycorrhizal fungi, but research shows that it generally depresses root

colonisation and alters the benefit of the symbiosis. In PQNK, moist, aerated soil permits mycorrhizal fungi and the wider microbial community to remain active around living roots. This should improve access to phosphorus and micronutrients.

Biological regulation can remove the routine need for pesticide. Flooding, excessive soluble nitrogen, and a simplified habitat can favour soft vegetative growth while weakening the predator network. In the PQNK field record described here, balanced plant growth and the restored field food web kept pests below the intervention threshold, so no pesticide was applied.

FIELD ADOPTION ACROSS COUNTRIES

The 2011 publication attracted interest from researchers and farmers outside Pakistan. After the founding trial was reported, the PQNK concept of growing rice on beds in moist soil, rather than in flooded, saturated soil, was taken up in Pakistan, South Korea, Israel, China, Thailand, India, and Nigeria, as Pedaver records show. Their methods are not all PQNK: how the beds are formed, watered and managed differs from country to country. PQNK beds are watered by a flow down the furrows only when the ball test shows the soil needs it.

South Korea. Pedaver records describe raised-bed rice adapted to the Korean climate and Japonica varieties, with competitive yields and substantially reduced water use. A Pedaver video record from South Korea shows nursery seedlings transplanted by machine into dry soil on raised beds, with furrows between them and no standing water, the beds formed by tractor just ahead of the transplanter.



Rice seedlings transplanted by machine into dry soil on raised beds in South Korea, with furrows between the beds and no standing water (Pedaver video record).

Israel. Rice has been grown without standing water under water-scarce conditions, with the beds watered through drip lines rather than flooded. Israeli drip-irrigation work has since taken this method to rice farms in Europe and Asia, showing that rice can be grown where conventional paddy would be impractical.

Pakistan. Farmer-led adoption has extended PQNK direct-seeded rice beyond the original Punjab trial. Pedaver records include fields in Punjab and Sindh, supported by photographs, videos, and farmer accounts.

THE METHANE DIMENSION: PQNK RICE AND CLIMATE

Flooded rice is a major agricultural source of methane. Prolonged saturation creates anaerobic conditions in which methanogenic microorganisms decompose organic material, and much of the methane reaches the atmosphere through the rice plant. Global methane assessments attribute roughly 8 percent of human-caused methane emissions to flooded rice.

PQNK rice produces far less methane than continuously flooded paddy because its raised beds stay moist and aerated rather than submerged, so the anaerobic conditions in which paddy methane forms do not develop. Evidence from other rice systems shows that aerobic periods without standing water sharply reduce methane.

Converting continuously flooded paddy to PQNK beds therefore removes the main source of rice methane wherever the change is made, which makes PQNK rice a significant contribution to agricultural methane mitigation.

PQNK rice addresses yield, water, methane, and soil condition through one root-zone design. The founding trial, with a hybrid coarse rice, more than tripled the regional yield while using about 70 percent less water. The later mature Basmati record yields at least 20 percent more than conventional Basmati, with no purchased inputs and about 8 percent of the water of its flooded-rice reference. Aerated beds remove the persistent conditions that generate most paddy methane, while permanent roots and mulch rebuild soil structure.

PQNK RICE IN PRACTICE: WHAT FARMERS REPORT

The peer-reviewed trial provides measured evidence; the Pedaver archive shows how later PQNK rice fields were implemented. It contains photographs, videos, and farmer accounts covering land preparation, establishment, water management, crop development, and harvest.

Farmers commonly report lower irrigation labour because the field is not kept flooded. Precision establishment also produces a more uniform stand and maturity pattern when the planter is correctly calibrated and operated. Farmers describe larger, better-filled panicles, and side by side with conventional Basmati their PQNK fields yield at least 20 percent more.

After harvest, PQNK fields retain roots and surface residue rather than being puddled for the next crop. Farmers report darker soil, more visible earthworms, and improved aggregation as the system matures. These are the visible signs of soil structure and

biological activity recovering.

THE INSTITUTIONAL RESPONSE: WHY PQNK RICE HAS NOT SCALED FASTER

The important question is why a published result with large implications for yield and water has not yet been taken up broadly by research and extension institutions. Chapter Twelve considered the wider institutional problem. Rice adds a practical constraint: research programmes, machinery, irrigation layouts, and performance standards have largely been designed around puddled and flooded production.

A non-flooded raised-bed system asks institutions to adopt a different production environment, not merely a different input within the existing paddy system. It works as a complete protocol: root-zone preparation, bed geometry, establishment machinery, moisture management, and mulch. Partial adoption produces a different system and a misleading result.

Farmer-to-farmer communication and the Pedaver knowledge network have carried most of the adoption recorded so far. National scaling requires trained Production Managers, precision machinery, and transitional risk protection for farmers who follow the full PQNK protocol.



PQNK Rice field - 2026 harvest
https://youtu.be/WFErthS3rT4

WHAT THIS CHAPTER HAS ESTABLISHED

Rice provides PQNK's strongest proof of concept because conventional paddy conflicts with all four governing rules. The founding raised-bed trial, with a hybrid coarse rice, more than tripled the regional yield while using about 70 percent less water and was published in a peer-reviewed journal in 2011. Later mature PQNK field records of Basmati document direct seeding into permanent mulched beds, no puddling or standing water, no purchased inputs, yields at least 20 percent above conventional Basmati, and crop-life water use of about 8 percent of the flooded-rice reference. The two records involve different varieties and inputs; they are complementary, but they are not one combined trial.

The founding evidence is published, and the mature system is established on farmers' fields. The next chapter turns to wheat, a staple of world food security, and Pakistan's NARC trial that recorded an average of 29 tillers per plant, a seed rate of 4.24 kg per acre, and production costs of Rs 11,660 per acre compared with Rs 26,236 for the comparison system.